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I.C JOSHI (IVTH EDITION)

CHOOSE THE CORRECT ANSWER ONLY

ALL PRINTS SHOULD BE TAKEN IN LEGAL SIZE ONLY OR CHANGE PRINTING SETTING TO A4 SIZE

1. ATMOSPHERE

001) Lowest layer of atmosphere is

a) Troposphere

b) Tropopause

c) Stratosphere

002) Height of Tropopause at equator is

a) 10-12 km

b) 16-18 km

c) 12-14 km

003) Height of Tropopause at Poles is

a) 12-14 km

b) 12-13 km

c) 08-10 km

004) Higher the surface temperature..... would be the tropopause

a) Higher

b) Lower

c) Same

005) Height of tropopause.....

a) Is constant

b) Varies with altitude

C) Varies with Latitude

006) Above 8 km the lower temperatures are over

a) Equator

b) Mid Latitudes

c) Poles

007) Atmosphere is heated by

a) Solar Radiation

b) Heat from earth surface

c) From above

008) Tropos means.....

a) Turning

b) Under current

c) Convection

009) CO₂ and H₂O are also called

a) Green House Gases

b) Rare Earth Gases

010) Troposphere is generally

- a) Stable
- b) Unstable**
- c) *Neutral*

011) Stratosphere is

- a) Unstable
- b) *Neutral***
- c) Stable**

012) Tropopause is discontinuous at about

- a) 30°lat**
- b) 40°lat**
- c) 60°lat

013) Most of atmospheric mass is contained in ...

- a) Troposphere**
- b) Stratosphere
- c) Heterosphere

014) Stratosphere extends from Tropopause to

- a) 50 km**
- b) 60 km
- c) 40 km

015) The middle atmosphere layer characterised by temperature inversion and stability...

- a) Troposphere
- b) Tropopause
- c) Stratosphere**

016) Mother of *pearl clouds occur* in.....

- a) Mesosphere
- b) Thermosphere
- c) Stratosphere**

017) The temperature in *ISA at 17 km* is ..

- a) -56.5°C**
- b) -65.5° C
- c) -35.5° C

018) *By weight*, approximate ratio of O₂ to N₂ in the atmosphere is

- a) 1:3**
- b) 1:4
- c) 1:5

019) *By volume*, the approximate ratio of O₂ to N₂ in the atmosphere is

- a) 1:3
- b) 1:4**
- c) 1:5

020) By volume, the proportion CO₂ in the atmosphere is

- a) 3%
- b) 0.3%
- c) 0.03%**

021) In ISA, the mean sea level temperature is

- a) 15° C**
- b) 10° C
- c) 25°C

022) Maximum concentration of ozone is at a height of.....

- a) 10-15 km
- b) 20-25 km**
- c) 30-35 km

023) Additional oxygen is needed while flying above

- a) 5000 ft
- b) 7000 ft
- c) 10000 ft**

024) CO₂ and H₂O keep the atmosphere

- a) Warm**
- b) Cold
- c) Have no effect

025) *Noctilucent clouds* occur in

- a) Thermosphere
- b) Mesosphere**
- c) Stratosphere

026) Temperature at 2 km is 05 °C what is ISA deviation. Hint : (Actual-ISA)

- a) -05°C
- b) -02° C
- c) -03°C**

027) Pressure at MSL is 1002.25 hPa. Find the ISA deviation. Hint : (Actual-ISA)

- a) -11 hPa**
- b) 10 hPa
- c) 12 hPa

028) In actual atmosphere temp, at 19 km is -60°C . How much it differs from ISA deviation?

- a) -4.5°C
- b) -05.5°C
- c) **-03.5°C**

029) Nacreous clouds occur in

- a) Thermosphere
- b) Mesosphere
- c) **Upper Stratosphere**

030) The atmosphere up to 80 km has a nearly similar composition and is called the Homosphere. Its uniform composition is due to.

- a) Pressure
- b) **Gravitation of earth**
- c) Mixing due to Turbulence

031) Half of the atmosphere, air mass is contained..... below

- a) **20,000 ft**
- b) 15,000 ft
- c) 10,000 ft

032) In jet standard atmosphere the Lapse rate is

- a) **$2^{\circ}\text{C}/1000\text{ ft}$**
- b) $2^{\circ}\text{C}/\text{km}$
- c) $5^{\circ}\text{C}/\text{km}$

033) The rate of fall of temperatures with height, called

- a) Isothermal rate
- b) Inversion rate
- c) **Lapse rate**

034) In actual atmosphere the lapse rate could

- a) **assume any value**
- b) fall up to 8 km
- c) rise up to 50 km

035) Tropical Tropopause extends from the equator to lat 35° - 45° . over India it is at

- a) 20 - 21 km
- b) 14 - 15 km
- c) **16 - 16.5 km**

036) Lapse rate in the troposphere is produce by..... and in the stratosphere by.....

- a) evaporation; condensation
- b) **Rising air; solar radiation**
- c) terrestrial radiation; solar radiation; convection
- d) solar radiation; convection

037) Most of the water vapour in the atmosphere is confined upto

- a) Stratosphere
- b) 30,000 ft**
- c) mid troposphere
- d) lower troposphere

038) Negative lapse rate of temperature is

- a) Isothermal rate
- b) temperature rise with lowering height
- c) temperature rise with increase in height (INVERSION)**
- d) temperature fall with height

039) In ICAO ISA the atmosphere is assumed to be isothermal

- a) In stratosphere
- b) 11 to 16 km
- c) 11 to 20 km
- d) 11 to 32 km**

040) One of the Characteristics of our atmosphere is

- a) poor conductor of heat and electricity**
- b) equator is warmer than poles above 10 km
- c) lapse rate in the stratosphere is positive
- d) density is constant above 8 km

041) Heat transfer in the atmosphere is maximum due to

- a) convection
- b) radiation
- c) sensible heat**
- d) latent heat**

042) The knowledge of the height of tropopause is important for a pilot because

- a) weather is mainly confined up to this level.**
- b) clouds rarely reach up to this height due to jetstreams.
- c) stratosphere starts at this height where all solar radiation is absorbed.

043) In ISA atmosphere the tropopause occurs at a height of

- a) 8 - 10 km
- b) 11 km**
- c) 16 - 18 km

044) Lapse rate in the troposphere is due to..... and in the Stratosphere is due to

- a) Conduction, convection
- b) terrestrial radiation, solar radiation
- c) Rising air, solar radiation**
- d) Water vapour, Ozone.

045) There is reversal of temperature in the atmosphere at 8 km because

- a) Lapse rate at poles is always higher than at equator
- b) Lapse rate at equator is always higher than at poles
- c) Lapse rate at equator is always higher than at poles
- d) *Lapse rate reverses at poles and becomes negative***

2. ATMOSPHERIC PRESSURE

046) Winds in a low pressure

- a) *Converge***
- b) Diverge
- c) Go straight

047) Low pressure is associated with

- a) Good Weather
- b) *Bad Weather***
- c) None

048) In a high pressure area winds are

- a) Normal
- b) Strong
- c) *Weak***

049) Flying from Low to High an altimeter would read

- a) Over
- b) *Under***
- c) constant

050) Isallobars are lines of equal

- a) Pressure
- b) Temperature Tendency
- c) *Pressure Tendency***

051) What kind of a barometer is an altimeter?

- a) *Aneroid***
- b) Mercury
- c) Alcohol

052) A region between two Lows and Two Highs is

- a) Depression
- b) Secondary Low
- c) *Col***

053) Bad weather and better visibility is associate with

- a) High
- b) *Low***
- c) Col

054) The relationship between height and pressure is made use in construction of

- a) *Altimeter***
- b) ASI
- c) V S I

055) Altimeter always measure the height of aircraft above

- a) MSL
- b) datum level of 1013.2 hPa
- c) **datum level at which its sub-scale is set**

056) Two aircraft flying at the same indicated altitude with their altimeter set to 1013.2 hPa. One is flying over cold air mass and other over warm air mass. Which of the two has greater altitude?

- a) **Ac flying over warm air mass**
- b) Ac flying over cold air mass

057) The rate of fall of pressure with height in a warm air mass compared to cold air mass will be

- a) Same
- b) More
- c) **Less**

058) An increase of 1000 ft at msl is associated with decrease of pressure of

- a) 100 hPa
- b) 1000 hPa
- c) 3 hPa
- d) **33 hPa**

059) Lines drawn through places of equal pressure are known as

- a) **Isobars**
- b) Isotherms
- c) Isogonal
- d) Isoclinal

060) Which is true

- a) Trough has frontal characteristics
- b) **At trough winds back in N - hemisphere**
- c) At trough winds veer in N- hemisphere

061) Semi Diurnal pressure changes are most pronounced in

- a) Polar region
- b) Middle latitudes
- c) **Tropics**

062) Flying from Delhi to Calcutta at constant indicated altitude but, experiencing a drift to Starboard. The actual altitude will be (Vis-a-vis) indicated altitude

- a) **Lower**
- b) Same
- c) Higher

063) In the Southern Hemisphere, around a Low Pressure Area wind blows

- a) **In clockwise direction**
- b) In anticlockwise direction
- c) Across isobars towards the centre

064) Altimeter of a/c on ground reads aerodrome elevation, its sub-scale is set to

- a) **QNH**
- b) QNE
- c) QFF
- d) QFE

065) Instrument for recording pressure is called

- a) Anemograph
- b) Barometer**
- c) Hygrograph

066) Poor visibility is associated with

- a) High**
- b) Low
- c) Col

067) On either side, perpendicular to the pressures rise

- a) Trough**
- b) Ridge
- c) Low

068) Fall of pressure with height is more rapid in

- a) Cold areas**
- b) Warm areas
- c) Humid areas

069) 300 hPa in ISA corresponds to the level

- a) 20,000 ft
- b) 30,000 ft**
- c) 35,000 ft

070) 18,000 ft height in ISA corresponds to level

- a) 700 hPa
- b) 200 hPa
- c) 500 hPa**

080) 200 hPa in ISA corresponds to the level

- a) 20,000ft
- b) 30,000 ft
- c) 40,000 ft**

081) 24,000 ft height in ISA corresponds to level

- a) 400 hPa**
- b) 500 hPa
- c) 300 hPa

082) 700 hPa in ISA corresponds to the level

- a) 20,000ft
- b) 10,000 ft**
- c) 18,000 ft

083) 40,000 ft height in ISA corresponds to level

- a) 400 hPa
- b) 500 hPa
- c) 200 hPa**

084) 850 hPa in ISA corresponds to the level

- a) 7,000ft
- b) 5,000 ft**
- c) 10,000 ft

085) Atmospheric pressure is due to

- a) wind
- b) temperature
- c) gravity
- d) density**

086) An aircraft is gaining altitude, inspite of altimeter reading constant altitude. why?

- a) Standard pressure has risen
- b) Flying towards High**
- c) Flying Towards Low
- d) Temperature has decreased

087) A contour of 9160 **m** can be expected on a constant pressure chart for pressure level

- a) 500 hpa
- b) 400 hpa
- c) 300 hpa**
- d) 200 hpa

088) In contour chart of 300 hpa, isohypse (contours) are drawn at interval of

- a) 20 gpm
- b) 40 gpm
- c) 60 gpm
- d) 80 gpm**

089) In constant pressure chart of 500 hpa, isohypse are drawn at interval of

- a) 20 gpm
- b) 40 gpm**
- c) 60 gpm
- d) 80 gpm

090) QNH of an aerodrome 160 m AMSL is 1005 hpa. QFE ? (Assuming 1 hpa = 8 m)

- a) 1010 hpa
- b) 975 hpa**
- c) 1005 hpa
- d) 990 hpa

091) Steep Pressure gradient would mean

- a) Contours far apart and weak wind
- b) Contours far apart and strong wind
- c) Isobars far apart and temperature low
- d) Isobars closely packed and strong wind**

092) What type of inversion occurs when a stable layer lies in a high pressure area

- a) Negative
- b) Radiation
- c) Subsidence**
- d) Airmass

093) Which of the following would cause true altitude to increase when altimeter indicates constant altitude

- a) Warm/Low
- b) Cold/Low
- c) Hot/High**
- d) Cool/Low

094) The movement of wind in relation to a cyclone is

- a) Descending and subsiding
- b) *Ascending and converging***
- c) Descending and cooling
- d) Ascending and diverging

095) An aerodrome is at the mean sea level. Its QNH is 1014.0 hpa. Its QFF will be

- a) *1014.0 hpa***
- b) 1013.25 hpa
- c) Difficult to tell
- d) More than QNH

3. TEMPERATURE

096) Diurnal variation of temperature is greatest when wind is

- a) *calm***
- b) light
- c) strong

097) Diurnal variation of temperature is maximum over

- a) forest
- b) ocean
- c) *land***

098) On a clear day the amount of solar radiation received by earth surface is

- a) 3/4 th
- b) 30%
- c) *5/6 th***

099) ALBEDO is

- a) Radiation received by earth
- b) Amount of heat
- c) *Reflecting power of earth***

100) During Day the ambient temperature is ... than ground

- a) *Lower***
- b) Higher
- c) Same

101) Diurnal variation of temperature over ocean is ...

- a) More than land
- b) Above 3°C
- c) *Less than 1°C***

102) At a coast station the diurnal variation of temperature, depends on ..

- a) *Wind direction***
- b) Wind speed
- c) Radiation

103) Snow surface reflects about..... % of solar radiation.

- a) 75%
- b) 80%**
- c) 90%

104) Amount of Solar radiation received per unit area is

- a) Insolation**
- b) Convection
- c) Radiation

105) Solar radiation received by the earth is

- a) Long Wave
- b) Albedo
- c) Shortwave**

106) Rise in temperature of a surface is proportional to its specific heat

- a) Directly
- b) Indirectly**

107) Specific heat of land is..... than that of water

- a) Lower**
- b) Same
- c) Higher

108) Minimum temperature is reached at ...

- a) sunrise
- b) midnight
- c) 1/2 - 1 hour after dawn**

109) An air parcel is lifted till it gets saturated. The temperature attained by it is called

- a) Potential temperature
- b) Dew Point**
- c) Wet bulb

110) Cloudy nights are.....

- a) cold
- b) normal
- c) warm**

111) Water vapour is transparent to terrestrial radiation

- a) completely
- b) partially**
- c) indifferent

112) Higher the temperature would be the wavelength of emitted radiation

- a) longer
- b) shorter**

113) Air is a bad conductor of heat. A parcel of air can therefore be regarded as insulated from the environment

- a) False
- b) True**

114) Warmer the earth..... will be the Nocturnal radiation

- a) **intense**
- b) weaker
- c) moderate

115) Heat is the..... of the KE of all molecules and atoms of a substance

- a) **sum total**
- b) average

116) The solar radiation consists of about 46 %

- a) UV
- b) **IR**
- c) Visible

117) The total energy radiated by a black body is proportional to its temperature (T)

- a) T^2
- b) T^3
- c) **T^4**

118) Intense radiation are emitted by

- a) **Hot bodies**
- b) Cold bodies
- c) Stars

119) The wavelength of most intense radiation is inversely proportional to its.....

- a) **Absolute temperature**
- b) Humidity
- c) Albedo

120) Hot bodies (like sun) radiate

- a) **Short Waves**
- b) Long Waves
- c) Both

121) The flow of heat from earth surface is 77% by

- a) Sensible Heat
- b) **Latent Heat**

122) $-40^{\circ}\text{C} = -40^{\circ}\text{F}$

- a) **True**
- b) False

123) Surface Temperature is recorded at a height of above ground

- a) 1.5 m
- b) **1.25m**
- c) 2 m

124) The door of Stevenson's screen should open

- a) **opposite to sun**
- b) into sun
- c) any direction

125) The liquid used in Minimum Thermometer is

- a) mercury
- b) alcohol**
- c) spirit

126) Freezing point of water is

- a) 0° F
- b) 12° F
- c) 22° F
- d) 32° F**

127) Boiling point of water is

- a) 100° F
- b) 112° F
- c) 212° F**
- d) 312° F

128) Freezing point of water is

- a) 173° K
- b) 273° K**
- c) 373° K
- d) 473° K

129) Boiling point of water is

- a) 373° K**
- b) 273° K
- c) 173° K
- d) 312° K

130) Convert 68° F into Kelvin temperature

- a) 233° K
- b) 283° K
- c) 294° K
- d) 293° K**

131) Diurnal variation of temperature is least on a day when it is

- a) Clear
- b) Partly cloudy
- c) Cloudy
- d) Overcast**

132) A clear and calm night is cooler than a cloudy night, because nocturnal radiation

- a) escape through cloud
- b) are partly radiated back by clouds to earth**
- c) are fully absorbed by H₂O
- d) are fully prevented by clouds to escape

4. AIR DENSITY

133) Density is at poles than equator

- a) **Higher**
- b) Lower
- c) Same

134) Above 8 km density is at poles than at equator

- a) Higher
- b) **Lower**
- c) Same

135) The altitude in ISA at which air density is the same as the observed density is called

- a) **Density Altitude**
- b) ISA Density
- c) Real Density

136) Density is usually expressed as

- a) Kg/sq m
- b) **g/cu m**
- c) N/sq m

137) Higher density altitude means density

- a) Higher
- b) **Lower**
- c) Same

138) For given pressure and temperature moist air has density

- a) Higher
- b) **Lower**
- c) Same

139) Air is less denser in

- a) High Altitudes
- b) Warm Air
- c) High humidity
- d) **All these**

140) Density altitude may be defined as:

- a) The altitude in a standard atmosphere at which the prevailing pressure occurs.
- b) **The altitude in a standard atmosphere at which the prevailing density occurs.**
- c) Surface of constant atmospheric pressure related to standard atmosphere of 1013.2 hPa

141) If pressure increases the density altitude

- a) Increases
- b) Lowers
- c) **Remains the same**

142) For every 1°C change in temperature, density altitude differs by

- a) **33 ft**
- b) 100 ft
- c) **120 ft**
- d) 210 ft

5. HUMIDITY

- 142) The ratio in % between the amount of water vapour present in the air to the amount of water vapour that it can hold at the same temperature is
- a) Humidity
 - b) Relative humidity**
 - c) Dew point
- 143) The temperature to which air be cooled at constant pressure to become saturated, is called
- a) Wet bulb temperature
 - b) Dry bulb temperature
 - c) Dew point**
 - d) Humidity
- 144) Free air temperature, Wet bulb temperature and Dew point temperature are equal when
- a) Air temperature is 0°C
 - b) Relative humidity is 100%**
 - c) Air temperature is not below 0°C
- 145) On a rainy day compared to sunny day the length of runway required is
- a) More**
 - b) Less
 - c) Same
- 146) The spread (difference) between Free air temperature and Dew point temperature is when air is saturated
- a) Large
 - b) Least**
 - c) Same
- 147) The saturation vapour pressure over water is than the ice
- a) More**
 - b) Less
 - c) Same
- 147) As the temperature of the air increases, the amount of water vapour required to saturate it
- a) decreases
 - b) increases**
 - c) remains same
- 148) The actual amount of water vapour contained in a given volume of air at a given temperature is termed as ...
- a) Relative Humidity
 - b) Specific Humidity
 - c) Absolute Humidity**
- 149) Humidity Mixing Ratio when air is lifted adiabatically
- a) decreases
 - b) remains constant**
 - c) increases
- 150) It is the lowest temperature which air would attain by evaporating water into it to saturate it.
- a) Wet bulb temp**
 - b) Dry bulb temp
 - c) Dew point

6. WIND

151) In S hemisphere if an observer faces wind, low will be to his

- a) Right
- b) Left**

152) In N hemisphere due to rotation of earth winds are deflected to

- a) Left
- b) Right**

153) Local Winds follow Buys Ballot's law

- a) False**
- b) True

154) Coriolis force acts perpendicular to the of wind direction in N hemisphere

- a) Left
- b) Right**

155) Geostrophic wind is due to the balance between the forces

- a) Coriolis and Friction
- b) Pressure gradient and Cyclostrophic
- c) Pressure gradient and Coriolis**

156) Coriolis force is strongest at

- a) Mid latitudes
- b) Poles**
- c) Equator

157) Geostrophic rule breaks down at

- a) Mid latitudes
- b) Poles
- c) Equator**

158) Fohn winds are on the Leeward side of a mountain,

- a) Dry & Warm**
- b) Cold & Humid

159) The wind sliding down a hill during night is called wind.

- a) Fohn
- b) Anabatic
- c) Katabatic**

160) With the onset of sea breeze there is a in temperature and in RH.

- a) Fall/Rise**
- b) Rise/Fall
- c) Fall/Fall

161) Sea breeze sets in by and dies off at

- a) Night/Day
- b) Day/Night**
- c) Both Day and Night

162) If an aircraft in N-hemisphere flies from H to L it will experience

- a) **Starboard drift**
- b) Port drift

163) In N-Hemisphere if you experience Port drift , altimeter will read

- a) **Under**
- b) Over

164) Lines of constant wind speed drawn on weather charts are called

- a) Isobars
- b) **Isotachs**
- c) Isogons

165) Squall are distinguished from gusts by:

- a) Shorter duration
- b) **Longer duration**
- c) Lower wind speed

166) The thermal wind is:

- a) The wind that blows because of thermals
- b) The warm wind that blows down the hill on the leeward side
- c) **The wind which must be added vectorially to the lower level geostrophic wind to obtain the upper level geostrophic wind**

167) On a weather map where isobars are closely packed, the surface winds are likely to be

- a) Light and parallel to isobars
- b) **Strong and parallel to isobars**
- c) Strong and blowing across the isobars

168) Anabatic wind occurs

- a) At night
- b) Any time of day and night
- c) **During day**

169) Anabatic wind is stronger than katabatic

- a) True
- b) **False**

170) Katabatic wind is down slope cold wind due to nocturnal cooling

- a) **True**
- b) False

171) Katabatic wind occurs due to sinking of air down the hill slope

- a) **True**
- b) False

172) Anabatic wind occurs due to downward movement of air along valley

- a) True
- b) **False**

173) Sea breeze is stronger than land breeze

- a) **True**
- b) False

174) The wind blows clockwise around low in a N-hemisphere

- a) True
- b) **False**

175) The wind blows clockwise around low in S-hemisphere

- a) **True**
- b) False

176) The wind blows anticlockwise around low in N-hemisphere

- a) **True**
- b) False

177) The wind blows anticlockwise around low in S- hemisphere

- a) True
- b) **False**

178) The resultant wind that blows under the influence of pressure gradient force, geostrophic force and cyclostrophic force is called

- a) **Gradient wind**
- b) Geostrophic wind
- c) cyclostrophic wind

179) Due to friction, from day to night for an isobaric pattern (in N hemisphere) the Surface wind backs and weakens

- a) **True**
- b) False

180) The winds which spiral inward in a counter-clockwise direction in the N Hemisphere are associated with

- a) Turbulence
- b) High pressure area
- c) **Low pressure area**

181) Lower level wind 05010 kt, upper level wind 23005 kt, what is the thermal wind

- a) 05005 kt
- b) **23015 kt**
- c) 05015 kt

182) A change in wind direction from 310° to 020° is

- a) Backing
- b) **Veering**

183) A change from 270° to 250° is

- a) **Backing**
- b) Veering

184) Sudden change in wind speed from 10 kt to 30 kt and then to 15 kt is

- a) **Gust**
- b) Squall
- c) gale

185) Sudden change in wind speed from 10 kt to 30 kt for 2 - 3 minutes

- a) **Squall**
- b) Gust

186) A significant wind shear can be associated with TS or line squall

- a) False
- b) **True**

187) Cyclostrophic wind gives a good approximation of the 2000' wind in an intense tropical storm

- a) **True**
- b) False

188) Rotor clouds have extremely turbulent flying conditions

- a) False
- b) **True**

189) Friction causes winds to flow cross isobaric by over land and over sea

- a) **20°/10°**
- b) 20°/30°
- c) 40°/30°

190) If the S is warmer than N, level by level, from surface up to higher levels, then thewind will strengthen with height with no change in direction in N hemisphere

- a) Ely
- b) **Wly**
- c) Sly
- d) Nly

191) Gradient wind is of geostrophic wind in an anticyclone

- a) Under estimate
- b) Accurate
- c) **Over estimate**

192) Gale is

- a) persistent strong winds with mean speed 44 kt, associated with thunderstorm
- b) marked increase in wind speed lasting few minutes associated with CB or DS (dust storm)
- c) **persistent strong winds exceeding 33 kt, associated with depression**

193) In N hemisphere thermal wind is parallel to with low value to left

- a) Isobars
- b) **Isotherms**
- c) Isallobars

194) The inertial flow is

- a) cyclonic in both the Hemispheres
- b) **anticyclonic in both the Hemispheres**
- c) anticyclonic around an anticyclone

195) Upper level wind is 24025 kt, lower level wind is 16015 kt, the thermal wind is?

- a) 16010 kt
- b) **24040 kt**
- c) 24010 kt

7. VISIBILITY AND FOG

196) Fog is reported when visibility is reduced to

- a) ***Less than 1000 m***
- b) 1000m
- c) 1000 to 2000

197) RVR is reported when visibility falls below

- a) 500 m
- b) 1000 m
- c) ***1500 m***

198) Radiation fog occurs

- a) ***Over land***
- b) Over sea
- c) During day

199) When visibility reduces between 5000 m and 1000 m and RH is almost 100%, it is

- a) ***Mist***
- b) Haze
- c) Fog

200) Radiation Fog forms over N India during

- a) May to June
- b) ***Dec to Feb***
- d) Oct to Nov

201) Warm and moist air moving over a cold ground gives rise to:

- a) Thunder clouds
- b) ***Fog and stratus***
- c) Frontal clouds

202) Warm and moist air moving over a cold surface causes

- a) Radiation Fog
- b) ***Advection Fog***
- c) Frontal Fog

203) The radiation fog forms due to

- a) Heating of the earth during day
- b) ***Radiational cooling of earth at night***
- c) Advection of cold air

204) The radiation fog activity increases after the passage of a

- a) ***WD***
- b) Depression
- c) Col

205) Radiation fog is essentially a phenomena

- a) **Nocturnal**
- b) Dusk
- c) Day

206) The radiation fog forms over

- a) Water
- b) **Land**
- c) Both

207) The fog forms due to horizontal movement of warm moist air over cold surface

- a) Radiation
- b) **Advection**
- c) Frontal

208) Advection fog forms during

- a) night only
- b) day time only
- c) **any time of day and night**

209) For formation of Radiation fog

- a) There should be sufficient moisture in atmosphere, cloudy sky, nil wind
- b) **There should be sufficient moisture in atmosphere, clear sky, light wind.**
- c) There should be sufficient moisture in atmosphere, cloudy sky, strong wind

210) Instrument used for measuring visibility is called ...

- a) Visiometer
- b) **Transmissometer**
- c) Ceilometer

211) Advection fog forms

- a) over sea
- b) over Land
- c) **both over land and sea**

212) fog is cloud on ground

- a) stratocumulus
- b) **stratus**
- c) Nimbostratus

213) Frontal fog is more common with a

- a) Western Disturbance
- b) cyclone
- c) **Warm Front**
- d) Cold Front

214) The favourable pressure system for formation of fog is

- a) Lows and Cols
- b) High and Trough
- c) Lows and Ridges
- d) *Highs and Cols***

215) Thermal processes / temperature distribution which favours formation of fog is

- a) Isothermal
- b) Adiabatics
- c) *Inversion***
- d) SALR

8. VERTICAL MOTION AND CLOUDS

216) Drizzle occurs from

- a) CS
- b) *ST***
- c) NS
- d) CU

217) Altostratus (AS) is

- a) a) Low cloud of sheet type
- b) *A medium cloud of sheet type***
- c) A cloud of large vertical growth
- d) A high cloud of sheet type

218) Showers occurs from

- a) CU
- b) *CB***
- c) AS
- d) AC

219) Heavy icing is possible in

- a) CI
- b) CS
- c) ST
- d) *CB***

220) To avoid icing in cloudy conditions, a pilot is advised to fly through a cloud which shows an optical phenomena

- a) *Halo***
- b) Corona
- c) With multi-coloured clouds

221) Dark gray cloud giving continuous rain is called

- a) AS
- b) *NS***
- c) ST
- d) CB

222) A uniform layer of cloud resembling fog but not on the ground

- a) AS
- b) NS
- c) ***ST***

223) The clouds composed of ice crystals having feathery appearance

- a) ***CI***
- b) CS
- c) AS

225) Halo is associated with the cloud

- a) AC
- b) AS
- c) ***CS***
- d) CI

224) NS clouds occur

- a) At cold front
- b) ***At warm front***

226) Corona is associated with the cloud

- a) ***AS***
- b) AC
- c) CC
- d) CS

227) Lenticular clouds indicate presence of

- a) Warm Front
- b) Cold front
- c) ***Mountain waves***

228) The lowest level below which condensation trails will not form is

- a) ***Mintra Level***
- b) Drytra Level
- c) Maxtra Level

229) Rain falling from cloud but not reaching ground is

- a) Virgo
- b) ***Virga***
- c) Mirage

230) CB with distinct anvil is called

- a) Castellanus
- b) ***Capillatus***
- c) Uncinus

231) Cloud ceiling is the height of the cloud covering

- a) 3-4/8
- b) 8/8
- c) ***5/8 or more***

232) No condensation trails occur above

- a) ***Maxtra Level***
- b) Dytra Level

c) Mintra Level

233) Cloud of operational significance has base below m or below the highest minimum sector altitude, which is greater

- a) **1500**
- b) 2000
- c) 1000

234) AC cloud with cumuliform protuberances are indicative of

- a) Stability
- b) **Instability**
- c) Neutrality

235) Hail may be experienced under the anvil of a CB

- a) **True**
- b) False

9. ATMOSPHERE STABILITY AND INSTABILITY

236) DALR = 9.8 °C, ELR = 6.8 °C; atmosphere is

- a) **stable**
- b) unstable
- c) indifferent

237) SALR = 5.5 °C, ELR = 4.5 °C ; atmosphere is

- a) unstable
- b) **stable**
- c) indifferent

238) DALR > ELR > SALR

- a) **conditionally stable**
- b) latently stable
- c) potentially stable

239) Dry air is unstable when

- a) ELR=DALR
- b) **ELR>DALR**
- c) ELR<DALR

240) The saturated air is said to be unstable if

- a) SALR=ELR
- b) **SALR<ELR**
- c) SALR > ELR

241) If ELR = SALR = DALR the atmosphere is

- a) Stable
- b) Instable
- c) **Indifferent**

242) DALR means:

- a) ***The rate at which temperature of unsaturated parcel of air falls with height when made to ascend adiabatically.***
- b) The rate at which temp falls with height.
- c) The rate at which ascending parcel of saturated air cools.

243) Surface temp, is 30° C. Assuming DALR. what is the temperature at 2 km

- a) 18° C
- b) 10° C**
- c) 42° C

244) An Isothermal atmosphere is

- a) Stable**
- b) Unstable
- c) Neutral

245) If environmental lapse rate(ELR) is less than SALR, the atmosphere is said to be:

- a) Absolutely unstable
- b) Conditionally stable
- c) Absolutely stable**

246) DALR is approximately

- a) 5 ° C /km
- b) 15 ° C /km
- c) 10 ° C /km**

247) SALR at mean sea level is about

- a) 10 ° C/Km
- b) 5 ° C/Km**
- c) 5 ° F/Km

248) SALR approaches DALR

- a) at 0°C
- b) at -15°F
- c) at -40°C**

249) Dry air having a temperature of 35°C on surface when forced to rise adiabatically by 1 km would attain a temperature of

- a) 29°C
- b) 25°C**
- c) 45°C

250) Inversion in the atmosphere indicates

- a) Stability**
- b) Instability
- c) Neutrality

251) Inversion is Lapse Rate (LR)

- a) Positive
- b) Negative**
- c) Neutral

252) Environmental LR can be more than DALR

- a) **True**
- b) False

253) The process which to a large extent determines the vertical distribution of temperature in atmosphere is

- a) **Adiabatic**
- b) Isothermal
- c) Isentropic

254) Rise in temperature with height is

- a) **Inversion**
- b) Lapse
- c) Normal

255) Inversion is common in

- a) Post Monsoon
- b) Monsoon
- c) **Winters**

10. OPTICAL PHENOMENA

256) Aurora Australis occur in the

- a) **S hemisphere**
- b) N hemisphere
- c) Equator

257) Aurora Australis called Lights

- a) Northern
- b) **Southern**
- c) Temperate

258) Aurora Borealis occur in the

- a) S hemisphere
- b) **N hemisphere**
- c) Equator

259) Aurora Borealis are called Lights

- a) **Northern**
- b) Southern
- c) Temperate

260) Corona occur in..... clouds

- a) **AS**
- b) NS
- c) CS

261) Bishop's ring is due to the diffraction of light by fine particles of

- a) water
- b) **dust**
- c) ice

262) The radius of the Bishop's ring is about

- a) 32°
- b) 22°**
- c) 42°

263) Superior Mirage occurs in marked

- a) Lapse
- b) Isothermal
- c) *Inversion***

264) Inferior Mirage occurs when there is

- a) *Lapse***
- b) Isothermal
- c) Inversion

265) Corona are formed due to the of light

- a) Refraction
- b) *Diffraction***
- c) Scattering

266) Corona are formed due to light, passing through

- a) Mist only
- b) Fog only
- c) small water or ice particles only
- d) *Any one of all these***

267) Halo is produced by

- a) *Refraction***
- b) Diffraction
- c) Scattering

268) Halo is produced when light passes through

- a) water particles
- b) *ice crystals***
- c) both

269) Halo occur in the cloud

- a) AS
- b) NS
- c) *CS***

270) Halo is luminous ring of radius

- a) 32°
- b) 22°**
- c) 42°

271) Halo round the sun shows a pure clear on the outside

- a) *Red***
- b) Yellow
- c) Violet

272) Halo signifies predominance in the cloud of

- a) Supercooled water drops
- b) *Ice crystals***
- c) Both

273) The cloud which cause Halo has ... chances of ice accretion

- a) *negligible***
- b) maximum
- c) medium

274) Sometimes a halo with a radius of is observed, called Large Halo

- a) 32°
- b) 42°**
- c) 22°

275) Halo occurs from.... cloud

- a) Low
- b) Medium
- c) *High***

11. PRECIPITATION

276) When super cooled water drops and ice particles co-exist, the ice crystals grow at the expense of the water drops because

- a) Saturation vapour pressure over water drops is less than over the ice crystals
- b) *Saturation vapour pressure over the ice crystals is less than over water drops***
- c) The ice crystals convert into water drops

277) The clouds whose tops extend well above the freezing level are called

- a) Warm Clouds
- b) *Cold Clouds***
- c) Moderate Clouds

278) The clouds whose tops do not extend to the freezing level are called

- a) *Warm Clouds***
- b) Cold Clouds
- c) Moderate Clouds

279) Coalescence Theory explains occurrence of rainfall from the

- a) *Warm Clouds***
- b) Cold Clouds
- c) Both types of Clouds

280) Ice crystal Theory explains occurrence of rainfall from the

- a) Warm Clouds
- b) *Cold Clouds***
- c) Both types of Clouds

281) Giant Nucleus Theory explains occurrence of rainfall over

- a) **Maritime areas**
- b) Inland areas
- c) Hilly areas

282) Very heavy precipitation as showers over a short period is called

- a) Flash floods
- b) **Cloud Burst**
- c) Orographic Rain

283) Rain shadow area is on the of the mountain range

- a) Top
- b) Windward side
- c) **Leeward side**

284) Sleet is a mixture of

- a) Hall & Snow
- b) **Rain & Snow**
- c) Frozen Rain

285) Rainfall in the tropics is more in

- a) Winters
- b) **Summers**
- c) Post monsoon

286) Rainfall in the tropics is more in the

- a) Morning
- b) **Afternoon**
- c) Night

287) Rainfall in the temperate latitudes is more in

- a) **Winters**
- b) Summers
- c) Spring

288) Over J&K and western Himalayas Rainfall is more in

- a) **Winters**
- b) Summers
- c) Post monsoon

289) Rainfall over coastal areas is more in the

- a) Evening
- b) Afternoon
- c) **Night & early morning**

290) Areas to the of western Ghats of India are rain shadow areas

- a) W
- b) S
- c) **E**

291) A sudden rise in the level of rivers or streams causing floods is called

- a) Cloud Burst
- b) Catchments flooding
- c) **Flash Floods**

292) Artificial rain making is also termed as

- a) Simulation
- b) *Cloud seeding***
- c) Nucleation

293) Fog can be dispersed for a short period by artificial stimulation

- a) *True***
- b) False

294) Showery precipitation occurs from

- a) NS
- b) AC
- c) *CB***

295) A day is called Rainy day when rainfall in 24 hr ismm or more

- a) 1.5
- b) 2
- c) *2.5***

12. ICE ACCRETION

296) Hoar frost occurs on airframe in clear air when the temperature of airframe is

- a) *below the frost point***
- b) frost point
- c) just above the frost point

297) In clouds at temperatures below 0° C an aircraft may encounter icing of the type

- a) only Glazed
- b) only Rime
- c) *intermediate between these two***

298) Opaque Rime ice is

- a) *Light porous***
- b) Solid
- c) Mixture of porous and solid

299) Rime is formed by freezing of supercooled water droplets on airframe when aircraft is flying through clouds

- a) *Small***
- b) Large
- c) Medium

300) Glazed ice is formed by freezing of supercooled water droplets on airframe when aircraft is flying through clouds

- a) small
- b) *Large***
- c) medium

301) The ice poses serious aviation hazard

- a) Rime
- b) Hoar Frost
- c) **Glazed**

302) Airframe icing occurs below 0°C. Its probability of occurrence decreases progressively below -20 °C, as at lower temperatures the proportion of super-cooled water drops in a cloud

- a) Increases
- b) **Decreases**
- c) Does not change

303) CI, CS and CC clouds consist mostly ice crystals. Icing hazard is therefore

- a) Maximum
- b) Medium
- c) **Negligible**

304) AS, NS consist of supercooled water drops and ice crystals in varying proportion... icing is possible.

- a) Maximum
- b) **Light or moderate**
- c) Negligible

305) In AC clouds icing is likely in mountainous areas

- a) Severe
- b) Light to moderate
- c) **Negligible**

306) In TCU icing may range from light to severe type at least up to level.

- a) - 40 °C level
- b) 30 °C level
- c) **- 20 °C level**

307) In CB icing may range from light to severe type up to -20°C level. Below this temperature severe icing is

- a) **not significant**
- b) significant
- c) maximum

308) Liquid water content is an important factor in icing. As the maximum water concentration is around, maximum ice formation in clouds may also be expected around that level.

- a) -25° C level
- b) -20° C level
- c) **-15° C level**

309) Carburetor icing occurs when air from intake passes through a venturi (choke) and causes expansional cooling and vaporization of fuel. Serious icing can occur at extreme temperatures

- a) 13 °C
- b) **30°C to -10°C**
- c) 20 °C

310) occurs in a moist cloudless air on an aircraft surface having temp. below 0° C, due to sublimation of water vapour onto feathery ice crystals.

- a) Rime
- b) Glazed
- c) **Hoar Frost**

- 311) occurs in St, Sc, Ac, Cu, Ns at temperature -10 to -40° C and in Cb at temperature -20 to -40°C
- Rime**
 - Glazed
 - Hoar Frost
- 312) In clouds ... occurs when a wide range of water drop sizes are present at temperatures between 0°C and -40°C
- Rime
 - Glazed
 - Mixture of rime and clear ice**
- 313) occurs in AS, NS, SC and towering CU or CB between 0° C and -20° C, in warm front below 0° C, especially if the aircraft has rapidly descended from a colder region
- Glazed
 - Fume
 - Mixture of Rime and Clear ice**
- 314) When fog freezes on parked aircraft it produces...
- Hoar Frost**
 - Hoar Frost
 - Clear ice
- 315) Icing the stalling speed appreciably
- Decreases
 - Increases**
 - Does not increase/decrease

13. THUNDERSTORM

- 316) The condition necessary for the formation of a thunderstorm are:
- Steep lapse rate, strong winds
 - shallow lapse rate, adequate supply of moisture
 - Steps lapse rate ,adequate supply of moisture and trigger action.**
- 317) Hail is most likely to fall from a cloud
- Having layers
 - Composed of Ice crystals
 - Having strong vertical development**
- 318) Norwesters are
- The western disturbances which affect NW India
 - Severe thunderstorms which occur over NE India during hot weather period**
 - Severe thunderstorms which occur over Peninsula during hot weather period
- 319) Dust Storm usually occurs over NW India during
- Post-monsoon
 - Winter
 - Pre-Monsoon**
- 320) A 'mature' thunderstorm has
- updraft only
 - downdraft only
 - updrafts and downdrafts**

321) Aircraft icing is most favoured in the cloud which have temperatures ranging between

- a) - 20° C and - 40° C
- b) 0° C and - 20° C**
- c) below -40 ° C

322) A short duration, showery precipitation is associated with

- a) ST
- b) AS
- c) CB**

323) Hail is

- a) Solid precipitation which commonly occurs over the mountainous regions in winter.
- b) Frozen or partly frozen rain falling from sheet type of clouds
- c) Solid precipitation falling from a deep convective cloud**

324) The most hazardous cloud for aviation is

- a) CB**
- b) CU
- c) NS

325) The life of a Cb cell is usually

- a) 7 to 8 hrs
- b) 3 to 4 hrs
- c) 2 - 3 hr**

326) Generally the severest activity of a, heat type, TS is for

- a) 2 hrs
- b) 30 to 45 min**
- c) 3 to 4 hr

327) Norwesters occur during

- a) Jan-Feb
- b) Mar-May**
- c) June-Sep
- d) Oct - Dec

328) Norwesters occur during

- a) Winter
- b) Hot weather**
- c) Monsoon
- d) Post Monsoon

329) Norwesters affect

- a) N India
- b) Bengal, Bihar, Orissa and Assam**
- c) Central India

330) The trigger action may take place due to

- a) Clear night sky no wind
- b) Orographic lifting**
- c) high pressure

331) Norwesters normally occur during

- a) Mornings
- b) *Afternoons***
- c) Nights

332) Norwesters originates over

- a) *Chota - Nagpur hills***
- b) Deccan Plateau
- c) Khasi hills

333) Andhi (blinding storms) occur generally over

- a) S India
- b) *N India***
- c) NE India

334) Wind speed in Light DS is

- a) 25 km
- b) 30 kt
- c) *up to 21 kt***

335) The diameter of Microburst is

- a) *less than 4 km***
- b) less than 2 km
- c) less than 6 km

336) The diameter of and Macroburst

- a) < 4 km
- b) *4 km or more***
- c) > 8 km

337) For detecting precipitation a Radar wavelength in the range ... is suitable

- a) *30 to 200 mm***
- b) 400-500 mm
- c) 600-700 mm

338) For airborne radars wavelength generally used

- a) 20 mm
- b) 40 mm
- c) *60 mm***

339) The wavelength of TS detection X band radar is

- a) 10 mm
- b) 20 mm
- c) *30 mm***

340) The wavelength of storm detection S band radar is

- a) 50 mm
- b) *100 mm***
- c) 200 mm

341) Over plains TS mostly occur during the

- a) *afternoon***
- b) night
- c) early morning

- 342) Over valley and foot hills TS generally occur during
- a) afternoon
 - b) *night & early morning***
- 343) Over the sea TS are more frequent
- a) afternoon
 - b) *night***
 - c) early morning
- 344) The life of Mesoscale Convective Complex TS is
- a) 2-3 hr
 - b) 3-4 hr
 - c) *6 to 24 hr***
- 345) Loud peals of thunder, frequent flashes of lightning, moderate or heavy showers accompanied by light hail with maximum wind speed 15-40 kt is classified as
- a) Light TS
 - b) *Moderate TS***
 - c) Severe TS
- 346) For a severe TS one of the requirements is strong wind shear
- a) Horizontal
 - b) *Vertical***
 - c) Slant
- 347) Severe TS cells are tilted
- a) *in vertical***
 - b) to the South
 - c) to the North
- 348) When flying through on active TS, lightning strikes are mostly likely
- a) Above 5000 ft and under the anvil
 - b) In the clear air below the cloud in rain
 - c) *In the temperature band between +10 C and -10 C***
 - d) At or above 10000 ft
- 349) Hazards of the mature stage of TS Cell include lighting, turbulence and
- a) Microburst, wind-shear and anvil
 - b) *Icing, microburst and WS***
 - c) Icing, drizzle and microburst
 - d) WS, hail and fog
- 350) Hail grows by
- a) freezing as it leaves the cloud
 - b) up and down forces in CU cloud
 - c) collision with ice crystals
 - d) *collision with supercooled water drops***

14. AIR MASSES, FRONTS AND WESTERN DISTURBANCES

351) The air mass which originates at sea in low Latitudes is called

- a) ***Polar maritime***
- b) Tropical continental
- c) Tropical maritime

352) The air mass which originates over equatorial region is

- a) Warm & dry
- b) ***Warm & Moist***
- c) Cold & dry

353) If the advancing cold front is **colder** than the **cool** air mass of the warm front, the advancing cold front undercuts and lifts both the warm and cool air masses of the warm front. This is

- a) Warm Occlusion
- b) ***Cold Occlusion***

356) The airmass which originated over land area located in polar region:

- a) Warm & dry
- b) Warm & Moist
- c) ***Cold & dry***

357) If a warm air mass overtakes a cold air mass, it is called

- a) Cold Front
- b) ***Warm Front***
- c) Occluded Front

358) At warm front

- a) ***Warm air overtakes the cold air***
- b) Cold air undercuts the warm air

359) The conditions are always unstable at

- a) ***Cold front***
- b) Warm front

360) Line squall occurs about 100-300 km ahead of

- a) Warm front
- b) ***Cold front***

361) Precipitation occurs over a belt of 30 - 50 km on both side of front

- a) ***Cold front***
- b) Warm front

362) Cold front moves at the speed of a warm front moves

- a) Same
- b) ***Double***
- c) Half

363) Line Squalls occur of Cold front

- a) **Ahead**
- b) Behind
- c) At the

364) Fronts are associated with

- a) Tropical cyclone
- b) Monsoon Depression
- c) **Extratropical Cyclones**

365) CB, Roll-type clouds, SC, AC with embedded CB are associated

- a) **Cold front**
- b) Warm front

366) The Surface of discontinuity between the Polar Easterlies and the temperate Westerlies is called

- a) Equatorial Front
- b) Tropopause
- c) **Polar Front**

367) The air mass which originates from sea area located in lower Lat is

- a) Warm & Dry
- b) **Warm & Moist**
- c) Cold & Moist

368) WDs approach India as

- a) Cols
- b) **Occluded Fronts**
- c) Highs

369) Maximum WDs occur in

- a) Summers
- b) Post Monsoon
- c) **Winters**

370) Ahead of a warm front the surface wind

- a) Backs & weakens
- b) **Veers & strengthens**
- c) Backs & strengthens

371) On approach of a Warm Front temperature

- a) Fall
- b) **Rise**
- c) Remain same

372) CI, CS, AS, NS, ST in sequence are associated with the front

- a) **Warm**
- b) Cold
- c) Occluded

373) During the passage of a Cold Front winds

- a) **Suddenly become squally**
- b) Back and weaken
- c) Veer and are of moderate strength

374) Visibility is poor in a Warm Front

- a) Ahead
- b) Ahead & During**
- c) After & During

375) Fog occurs in Cold Front

- a) Ahead
- b) During
- c) After**

378) WD is a..... front

- a) Cold
- b) Warm
- c) Occluded**

379) FZRA and FZFG occur of a warm front

- a) Ahead**
- b) During
- c) After

380) Precipitation ceases after the passage of a front

- a) Cold
- b) Warm**
- c) Occluded

381) During the approach of a Warm Front wind

- a) Backs**
- b) Veers
- c) Does not change

382) A WD has two or more closed isobars, at 2 hPa interval, it is termed as

- a) Troughs in Westerlies
- b) Western Depression**
- c) Western Cyclone

383) A WD originate over

- a) Baluchistan
- b) Caspian sea
- c) Mediterranean, Caspian and Black Seas**

384) Induced lows develop to the of a WD

- a) N
- b) S**
- c) NE

15. JET STREAMS

385) The arbitrary lower limit of jet core velocity has been assigned by WMO as

- a) **60 kt**
- b) 60 m/s
- c) 70 m/s

386) Jet stream has

- a) one maxima
- b) **one or more maxima**
- c) only two maxima

387) The vertical wind shear in a Jet stream is about

- a) **5m/s/ km**
- b) 6m/s/km
- c) 8m/s/km

388) Compared to horizontal wind shear the vertical wind shear in a Jet stream is

- a) weaker
- b) **stronger**
- c) same

389) In a jet stream, the path of the maximum speed is known as

- a) Core
- b) **Axis**
- c) Jet streak

390) The wind speed along the axis of a jet stream is always

- a) Uniform
- b) **Not uniform**

391) Along the axis of a jet stream there are centres of high speed winds, these are called

- a) **Jet streaks**
- b) Core
- c) Axis

392) In a wavy jet the Jet streaks are located over or near the

- a) **Ridge**
- b) Trough
- c) Between Trough and Ridge

393) Sub-tropical Jet Stream (STJ) is

- a) **Westerly**
- b) Easterly
- c) Southerly

394) The normal position of Sub-tropical Jet Stream is

- a) 30° N
- b) **27° N**
- c) 35° N

395) The southern most position of STJ is in February is

- a) **22° N**
- b) 20° N
- c) 18° N

396) The STJ affects India from

- a) Jun to Jul
- b) **Oct to May**
- c) Aug to Sep

397) STJ has a layered structure. There are often two layers of maximum wind to the of jet core

- a) **S**
- b) N
- c) SW

398) The STJ strengthens

- a) Northwards
- b) Upstream
- c) **Downstream**

399) At and near the STJ the temperature gradient is very

- a) **Small**
- b) Large
- c) Moderate

400) Vertical wind shear in STJ is greater the core

- a) **above**
- b) below
- c) along

401) The TJ prevails over the Indian Peninsula from

- a) May to Jun
- b) Sep to Oct
- c) **Jun to Aug**

402) The TJ is located over the Indian Peninsula, approximately at

- a) **13° N**
- b) 17° N
- c) 18° N

403) The TJ is located over Indian, approximately at a height of

- a) **15 -16 km**
- b) 12 - 13 km
- c) 11 - 12 km

404) The TJ is strongest in

- a) **July - Aug**
- b) Sep - Oct
- c) June

405) In the TJ the wind shears are much than the STJ

- a) more
- b) **less**
- c) same

406) TJ is

- a) Westerly
- b) *Easterly***
- c) Southerly

407) A jet Stream can be recognized by

- a) High level dust
- b) High Pressure
- c) *Streaks of CI***
- d) Lenticular clouds

408) Flying at right angles to a jet stream with falling pressure you will experience

- a) *wind from left***
- b) Increasing head wind
- c) Increasing tail wind
- d) Wind from right

409) When and Where tropical jet stream occurs

- a) All year along equator
- b) In middle East in summers
- c) In winters over Russia
- d) *In summers over SE Asia and Central Africa***

17. MOUNTAIN WAVES

410) For mountain waves to form there should be flow of air across the ridge, generally withinof the perpendicular to the ridge.

- a) *30°***
- b) 45°
- c) 60°

411) For mountain waves to form the wind speed for small mountains should be atleast

- a) 15 m/s
- b) 10 m/s
- c) *7m/s***

412) For mountain waves to form the wind speed for large mountains should be atleast

- a) *15 m/s***
- b) 10 m/s
- c) 7m/s

413) For mountain waves to form the atmosphere should be up to the ridge, where air stream strikes the ridge.

- a) Unstable
- b) *Stable***
- c) Indifferent

414) For mountain waves to form the atmosphere should be at higher levels above the ridge

- a) *Unstable***
- b) Stable
- c) Indifferent

415) In Mountain waves the Rotor clouds form in

- a) Troughs
- b) *Ridges***
- c) Valley

416) Clear air turbulence is often encountered

- a) At the boundary of a jet stream
- b) In the wake of a passing airplane
- c) In the wake of a larger airplane at take off and landing
- d) *All of the above***

417) Most CAT occurs on the of a jet stream and in the vicinity of upper level frontal zones where temperature contrasts are strong.

- a) *Fringes***
- b) Within the core
- c) Axis

418) CAT is the bumpiness experienced by aircraft at high altitudes in either cloud-free conditions or in stratiform clouds

- a) *above 18,000 feet***
- b) below 18,000 feet
- c) below 28,000 feet

419) When approaching an area where mountain waves have been reported, a pilot should expect:

- a) Possible presence of roll clouds and lenticular clouds
- b) Intense updrafts and downdrafts on the lee side of the mountains
- c) Moderate to severe turbulence as far as 20 to 30 miles from the range on lee side
- d) *All of the above***

18. TROPICAL SYSTEMS

420) Wind speed in a tropical severe cyclone is:

- a) 27 -33 kt
- b) *48 - 63 kt***
- c) 17-27 kt

421) Fronts are characteristic of:

- a) Tropical cyclone
- b) *Extra-tropical depressions***
- c) Monsoon depressions

422) In a mature tropical cyclone, the eye area is characterised by:

- a) moderate winds and heavy rains
- b) hurricane force winds and squalls
- c) *calm wind, little clouding and practically no rainfall***

423) cyclonic storm cross tamil Nadu coast in :

- a) *Oct - Nov***
- b) Jul - Aug
- c) Feb - May

424) Cyclonic storms forms over Indian seas during

- a) SW monsoon
- b) NE monsoon
- c) ***Pre - Monsoon & Post Monsoon***

425) TRS occurs over Indian seas in

- a) Equatorial Region
- b) ***5° - 15°N***
- c) 0° - 5° N

426) In the Bay of Bengal during the months of Jul, august, Monsoon Depression form

- a) ***Over Head bay***
- b) Central Bay
- c) South Bay

427) In a Monsoon depression maximum weather occurs in

- a) SE sector
- b) NW sector
- c) ***SW sector***

428) During re-curvature maximum weather in a monsoon depression occurs in

- a) SE sector
- b) NW sector
- c) SW sector
- d) ***NE sector***

429) On whatever compass course the cyclone is approached, strong winds from the port indicate that the centre lies somewhere

- a) ***Ahead***
- b) Behind
- c) Port
- d) Starboard

430) The well developed extra tropical cyclonic storm is composed of two main frontal systems and an occluded front, which varies in extent:

- a) A stationary Front followed by a warm front
- b) ***A low with a warm front radiating out Southwards followed by a cold front.***
- c) A warm front and an occluded front
- d) None of the above

431) cyclonic storms are confined almost entirely to the

- a) Latitudes near Equator
- b) ***Tropic latitudes***
- c) Polar lat.
- d) Mid lat.

432) The weather expected in a well developed cyclonic storm is:

- a) Moderate Weather
- b) ***Stormy weather***
- c) Clear skies
- d) No wind or temperature change

433) After entering land the cyclones gradually die down

- a) ***True***
- b) False

434) The is the most dangerous part of the cyclone is

- a) **eye wall**
- b) eye
- c) outer storm area
- d) complete cyclone

435) No CS form

- a) At Poles
- b) **At Equator**
- c) At Lat 40 deg
- d) All these

436) Average Life of a tropical cyclone in India is

- a) 10 Days
- b) 6-8 Days
- c) **3 - 4 Days**

437) CS in India are mostly of intensity

- a) Severe
- b) Very Weak
- c) **Moderate**

438) Eye of a CS is surrounded by

- a) Shelf Clouds
- b) **Wall Clouds**
- c) Rotor Clouds
- d) Roll clouds

439) CS in India region are less intense because

- a) **They have a very short travel over the sea**
- b) Sea surface temp, are not high
- c) India is close to Equator

440) Surge is sudden strengthening of wind in the air mass

- a) **Same**
- b) Different
- c) Both

441) Surge and Shear Line are Features of monsoon

- a) **Vigorous**
- b) Weak
- c) Moderate

442) Peninsular discontinuity occurs along

- a) West Coast
- b) East Coast
- c) **Central peninsula**

443) Peninsular discontinuity occurs during

- a) Monsoon
- b) NE Monsoon
- c) **Pre Monsoon**

444) Surge is also

- a) Velocity Divergence
- b) *Velocity Convergence***
- c) Like ITCZ

445) In a depression the wind speed is

- a) < 17 kt
- b) 34 - 47 kt
- c) *17 - 27 kt***
- d) 48 - 63 kt

446) In a CS the wind speed is

- a) < 17 kt
- b) *34 - 47 kt***
- c) 17 - 27 kt
- d) 28 - 33 kt

447) A monsoon depression Slopes with height

- a) SE
- b) NW
- c) *SW***
- d) NE

448) Average life of a monsoon depression is

- a) 2 - 3 days
- b) 8 - 9 days
- c) *3 - 5 days***
- d) 10 - 12 days

449) Monsoon Depression moves faster over

- a) Sea
- b) *Plains***
- c) Hills
- d) Rivers

450) For formation of a CS sea surface temperature should be

- a) *26° - 27° C***
- b) 23° - 24° C
- c) 24° - 25° C
- d) 30° C

451) In extra tropical cyclone family there are 3 - 4 CS or Depressions, one to the of the other

- a) SE
- b) NW
- c) *SW***
- d) NE

452) In TRS the central pressure is about

- a) 1002 - 1010 hpa
- b) *1000 - 900 hpa***
- c) 1010 - 1020 hpa

453) TRS develop over

- a) *Sea***
- b) Land
- c) Islands

d) At equator

454) TRS develop over deg Lat

- a) 0 - 5
- b) 5 - 15**
- c) 25 - 30
- d) 30 - 40

455) In TRS Pressure gradient is very steep up to a radius of

- a) 30 Km
- b) 50 km
- c) 100 km
- d) 400 km**

456) Extra Tropical Storms originate in

- a) Equatorial Front
- b) Tropical Front
- c) Polar Front**

457) Extra - tropical storms generally moves in the direction

- a) S to E
- b) W to E**
- c) S to W
- d) E to W

458) A tropical Cyclone moves in a day

- a) 300 - 500 km**
- b) 500 - 700 km
- c) 700 - 900 km

459) A Tropical CS has outer storm area of about

- a) > 400 Km**
- b) > 700 Km
- c) > 900 Km

19. CLIMATOLOGY OF INDIA

460) During winters

- a) Advection fog occurs over northern & central part of India
- b) Radiation fog occurs in southern part of country
- c) Activity of Radiation fog increases after the passage of a WD over N India**

461) Low Temperatures and low humidity is the characteristic of

- a) Post Monsoon
- b) Hot weather
- c) Monsoon
- d) Winter months**

462) Hot weather period is

- a) Jan - Feb
- b) Mar - May**
- c) Jun - Sept
- d) Oct - Dec

463) During hot weather

- a) **WDs cause TS / DS over Punjab & Rajasthan**
- b) No WD affect N-parts of country
- c) Track of WD is southern most

464) The monsoon current over the West coast of India is

- a) **SWly**
- b) SEly
- c) NEly

465) Monsoon period is

- a) Jan to Feb
- b) March to May
- c) **June to Sep**
- d) Oct to Dec

466) The monsoon advances with

- a) Bay of Bengal current only
- b) Arabian Sea current only
- c) **Bay of Bengal and Arabian sea currents**

467) The rainfall over India during monsoon depends on

- a) Low over Pakistan
- b) Depression over Bay of Bengal
- c) **The position of the axis of Monsoon Trough**

468) Rainfall occurs all over the country during monsoon when

- a) **Axis of MT is in its normal position along Gangetic plains**
- b) Axis of MT is along Himalayas
- c) Depression has formed over Bay of Bengal

469) Break in monsoon occurs when the axis of monsoon trough is along the

- a) Gangetic plains
- b) **foot hills of Himalayas**
- c) Bay of Bengal

470) With a depression over the head Bay fair weather during monsoon occurs over

- a) **Assam**
- b) W Bengal
- c) Orissa

471) Monsoon is also called

- a) NE monsoon
- b) **SW monsoon**
- c) SE monsoon

472) Post monsoon weather period is

- a) Mar to May
- b) Jun to Sept
- c) **Oct to Nov**
- d) Jan to Feb

473) During post monsoon, pressure gradient over India is

- a) **weak**
- b) steep
- c) same as winters

474) During monsoon period, low pressure lies over

- a) Bay of Bengal
- b) NE India
- c) Central India
- d) **Pakistan**

475) During vigorous monsoon period the pressure gradient over west coast is

- a) Weak
- b) **Steep**
- c) Normal

476) An aircraft flying in Monsoon season from Chennai to Kolkata at 14 km will experience winds

- a) **Easterly**
- b) Westerly
- c) Nly
- d) Southerly

477) An aircraft flying in Winter season from Chennai to Kolkata at 12 km will experience winds

- a) Ely
- b) **Wly**
- c) Nly
- d) Sly

478) An aircraft flying in Pre monsoon season from Delhi to Kolkata at 10 km will experience winds

- a) Easterly
- b) **Westerly**
- c) Northerly
- d) Southerly

479) An aircraft flying in Monsoon season from Mumbai to Ahmedabad at 03 km will experience winds

- a) SEly
- b) **SWly**
- c) Nly
- d) NWly

480) An aircraft flying in winter season from Delhi to Kolkata at 06 km will experience winds

- a) SEly
- b) SWly
- c) Nly
- d) **NW - Wly**

481) An aircraft flying in winter season from Delhi to Kolkata at 06 km will experience

- a) **Port drift**
- b) Starboard drift
- c) Tail wind
- d) Head win

482) An aircraft flying in winter season from Kolkata to Nagpur at 02 km will experience

- a) **Port drift**
- b) Starboard drift

- c) Tail wind
- d) Head wind

483) During break monsoon the pressures all over the country

- a) *Rise***
- b) Fall
- c) Do not change

484) During break monsoon sometimes the surface winds over East UP and Bihar are

- a) *very strong***
- b) weak
- c) normal

485) Mid tropospheric cyclone occurs during

- a) Pre monsoon
- b) Winters
- c) Post monsoon
- d) *SW monsoon***

486) Mid tropospheric cyclone during monsoon form over

- a) Orissa
- b) Punjab
- c) *Gujarat***
- d) Chennai

487) El Nino episode is applied by fishermen (of Ecuador and N peru) to a period of reduced fish catch due to suppression of upwelling

- a) *warm surface coastal current***
- b) Wly surface coastal current
- c) Cold surface coastal current

488) Maximum cyclones occur over India in

- a) Pre monsoon
- b) Winters
- c) *Post monsoon***
- d) SW monsoon

489) Tropical Jet stream occurs in India during

- a) Pre monsoon
- b) Winters
- c) Post monsoon
- d) *SW monsoon***

490) Pressure gradient over West Coast of India is steep during

- a) Pre monsoon
- b) Winters
- c) Post monsoon
- d) *SW monsoon***

491) During break in monsoon rain occurs

- a) *along foot hills***
- b) NW India
- c) S India

20. GENERAL CIRCULATION

- 492) The polewards moving air piles up in the subtropical regions and forms high pressure belt at the surface, called
- a) ***Subtropical high***
 - b) Polar High
 - c) Equatorial high
- 493) occurs over subtropical high
- a) advection
 - b) convection
 - c) ***subsidence***
- 494) The occurrence of large deserts near 30 N and 30 S are due to large scale
- a) ***subsidence***
 - b) convection
 - c) advection
- 495) A part of the sinking air over the subtropical highs flows towards the equator, turning west (in the northern hemisphere) due to the Coriolis force. This surface air is called
- a) Trade winds
 - b) Roaring forties
 - c) ***Doldrums***
- 496) The huge vertical circulations, one between the equator and 30 N and another between equator and 30 S, are called
- a) ***Hadley Cells***
 - b) Ferrel Cells
 - c) Polar cells
- 497) The descending branch of the Hadley cell marked by calm winds and high pressure at the surface are called
- a) Tropical Latitudes
 - b) Middle latitudes
 - c) ***Horse Latitudes***
- 498) The winds in the upper troposphere are westerly. These are known as
- a) Natural Westerlies
 - b) Steady Westerlies
 - c) ***Zonal Westerlies***
- 499) In the equatorial regions the upper tropospheric winds are
- a) Westerlies
 - b) ***Easterlies***
 - c) Zonal Westerlies
- 500) The disturbances of middle latitudes move from
- a) ***W to E***
 - b) E to W
 - c) N to S
 - d) S to N
- 501) The tropical disturbances which form in the equatorial low pressure belt move in a
- a) Easterly direction

- b) Westerly direction**
- c) Southerly direction

502) Tropical disturbance which reach the zone of transition in the upper level flow change course and begin to move in a

- a) Perpendicular direction
- b) Opposite direction**
- c) Southwards

503) Latitudinally, on the average there is radiation in the tropics than in the polar regions

- a) surplus**
- b) deficit
- c) balance

504) The systems like highs, lows, cyclonic circulation etc are associated with distinct types of weather. A study of the behaviour of these systems, known as

- a) Synoptic Meteorology**
- b) Climatology
- c) Physical Meteorology

505) Rising air creates calms or doldrums in the equatorial region.

- a) ITCZ
- b) Horse Latitudes
- c) Equatorial Doldrums**

506) Steady NE winds in the N hemisphere and SE winds in the S hemisphere are called.

- a) Easterly winds
- b) Trade Winds**
- c) Tropical Winds

21. METEOROLOGICAL SERVICES FOR AVIATION

507) For non-scheduled National Flights an advance notice (before ETD) is required to be given to AMOs

- a) 3 hr**
- b) 18-24 hr
- c) 6 hr

508) For non-scheduled National Flights an advance notice (before ETD) is required to be given to AMSs

- a) 3 hr
- b) 18-24 hr
- c) 6 hr**

509) WAFS (World Area Forecast System) provides high quality en-route forecasts of to Met Offices

- a) high quality Met en-route forecasts of winds and temperature**
- b) SIGMET
- c) TREND

510) IMD has Regional Met offices

- a) 4
- b) 5
- c) 6**

511) There are Aerodrome Met Office (Class I Met Offices)

- a) **14**
- b) 17
- c) 16

512) There are Class III Met Offices

- a) 42
- b) 52
- c) **62**

513) In AIREP the Met Information is contained in Section

- a) 1
- b) 2
- c) **3**

514) TAF are generally valid for

- a) 18 hr
- b) **12 hr**
- c) 9hr

515) The validity of Landing Forecast (TREND) is

- a) **1 hr**
- b) 2 hr
- c) 3 hr

516) Landing Forecast is append to

- a) METAR and SPECI
- b) **TAF**
- c) AIREP

517) Local Forecast is issued three times a day valid for next

- a) 18 hr
- b) **8hr**
- c) 9hr

518) Local Forecast covers an area

- a) **50 NM**
- b) 100 NM
- c) 150 NM

519) Prognostic Charts are issued by

- a) **RAFC**
- b) AMO (Class I Met Offices)
- c) MWO

520) Prognostic Charts are valid for

- a) 18 hr
- b) **12 hr**
- c) 9hr

521) S1G Wx Chart are issued for Flight Levels

- a) **below 460**
- b) 460
- c) above 460

522) CODAR is

- a) Radar Report
- b) Coded ARFOR
- c) ***Upper report from an aircraft (other than weather reconnaissance aircraft)***

523) WITEM is

- a) Actual upper winds
- b) ***Forecast upper wind and temperature***
- c) Actual temperature and upper winds

524) SIGMET is a notice of severe weather for

- a) actual
- b) expected
- c) ***both***

525) SIGMET is issued for aircraft

- a) ***in flight***
- b) on ground
- c) both

526) SIGMET is issued by

- a) RAFC
- b) AMO (Class I Met Offices)
- c) ***MWO***

527) SIGMET is valid for

- a) ***4 hr***
- b) 18 hr
- c) 6 hr

528) SIGMET is not issued for

- a) ***Rain***
- b) Volcanic Ash
- c) Severe Icing

529) AIRMET is SIGMET issued by MWO for the safety of

- a) ***low-level aircraft***
- b) high -level aircraft
- c) both

530) GAMET is an area forecast in abbreviated plain language for

- a) high -level aircraft
- b) ***low-level aircraft***
- c) both

531) The validity of Airfield warnings is not exceeding

- a) 3 hr
- b) ***4 hr***
- c) 6 hr

532) Airfield warning is issued for expected wind speed

- a) **30 kt**
- b) 24 kt
- c) 15 kt

533) Airfield warning is issued for wind direction of 20 kt changes by

- a) **45°**
- b) 30°
- c) 60°

534) Airfield Warning for gliders, light aircraft and helicopters is issued for expected wind speed

- a) 15 kt
- b) **17 kt**
- c) 30 kt

535) Wind Shear Warning is issued for the observed or expected wind shear above runway

- a) **up to 500 m**
- b) above 500 m
- c) 300 m

536) VOLMET is a MET information for aircraft

- a) on ground
- b) **in flight**
- c) both

537) VOLMET Radio Telephony Broadcast made on HF Channel from

- a) Mumbai only
- b) Kolkata only
- c) **both**

538) VOLMET consists of

- a) SIGMET
- b) TAP only
- c) **TAF and METAR**

539) In ROBEX the METAR/SPECI of international aerodromes and their alternates exchanged

- a) within India
- b) **outside India**

540) SIG Weather Charts are issued times a day

- a) Two
- b) Three
- c) **Four**

24. STATION MODEL

Ref : page no 221 for station model diagram in I.C joshi (4th edition)

541) The direction of surface wind is

- a) **NW**
- b) NE
- c) SE

542) The range of wind speed is

- a) 20-25 kt
- b) 16-22 kt
- c) **18-22 kt**

543) The type of medium cloud is

- a) **AC**
- b) AS
- c) AC & AS

544) Low clouds are

- a) CU
- b) SC
- c) **ST**

545) High clouds are

- a) CC
- b) **CI**
- c) CS

546) Height of base of low clouds is

- a) 200 m
- b) 250 m
- c) **100 to 199m**

547) Amount of lowest cloud is

- a) 2/8
- b) 4/8
- c) **3/8**

548) Wet bulb temperature range is

- a) **29.5 to 30.4° C**
- b) 29.1 to 30.4°
- c) 30.6 to 30.4°

549) Temperature 34° C is

- a) **Dry**
- b) Wet
- c) Dew Point

550) 962 represents

- a) QFE
- b) **QFF**
- c) QNH

551) Pressure value in hPa is

- a) 996
- b) **996.2**
- c) 1096.2

552) Pressure change of 1.4 hPa is in

- a) 6 hr
- b) 3hr
- c) **24 hr**

553) Past weather is

- a) **Moderate Light Rain**
- b) Continuous Rain
- c) Heavy Rain

554) Rainfall amount is

- a) 2 mm
- b) 1.5 mm
- c) **1.6 to 2.4 mm**

555) Rainfall reported is from

- a) **0300Z of yesterday to 0300Z of today**
- b) 0100 to 2400Z
- c) 0530 to 0830 1ST

556) Speed of Ship is

- a) 5 KMH
- b) 5 MPS
- c) **5 KT**

557) Direction of movement of Ship

- a) NW
- b) SW
- c) **NE**

558) Speed of Ship reported is averaged for

- a) **3 hr**
- b) 6 hr
- c) 12 hr

559) Surface visibility is

- a) 500 to <1000 m
- b) 1000 to 2000m
- c) **2000 to < 4000 m**

560) Weather reported is

- a) Rain at the time of observation
- b) **Rain during last one hour**
- c) Rain not at station but within 5 km

25. AVIATION WEATHER REPORTS and CODES OF METAR SPECI AND TREND

METAR VIDP 160230Z 30005KT 290V050 1500S 5000N R15/P1500U BR FEW020 FEW025CB SCT120 BKN300 32/29 Q1003 REFG TEMPO FM0330 22015G25KT 3000 +TSRA FEW010 SCT025CB BKN150 BECMG AT0415 27008KT CAVOK=

561) The METAR has been issued on day

- a) 15th
- b) **16th**
- c) 17th

562) The METAR has been issued at

- a) 0630 1ST
- b) 0230UTC**
- c) 0230 1ST

563) The surface wind speed is

- a) 2- 6 kt
- b) 3-7 kt**
- c) 4-6 kt

564) Wind direction is varying from

- a) 290 to 050°**
- b) 050 to 290°
- c) 200 to 050°

565) Visibility 1500 m is towards

- a) N
- b) S**
- c) All over the airport

566) Visibility towards N is

- a) 1500m
- b) 5000 m**
- c) 3000 m

567) Height of base of low clouds above station level

- a) 2000 ft**
- b) 2500 ft
- c) 2500 ft

568) Runway Visual range is

- a) 1500 m
- b) >1500 m**
- c) < 1500 m

569) Runway Visual range has

- a) Decreased
- b) Increased**
- c) Remained same

570) The poor visibility is due to the Present weather, which is

- a) Fog
- b) Brown Dust
- c) Mist**

571) Amount of lowest cloud is

- a) 1-2/8**
- b) 2-4/8
- c) 5-7/8

572) Amount of CB cloud is

- a) 1-2/8**
- b) 2-4/8

- c) $5\frac{7}{8}$

573) Height of base of CB is

- a) 2500 m
- b) 3000 ft
- c) **2500 ft**

574) The height of topmost layer of cloud is

- a) 3000 m
- b) 30000 m
- c) **30000 ft**

575) The landing forecast appended to METAR is valid for

- a) 1 hr
- b) **2hr**
- c) 3 hr

576) The wind in TREND from 0330 UTC is valid up to ... UTC

- a) 0400
- b) 0430
- c) **0415**

577) Expected visibility after 0415 UTC is

- a) 6000 m
- b) 10 km
- c) **$\geq 10\text{ km}$**

578) The group Q1003 in the METAR indicates

- a) QFE
- b) QFF
- c) **QNH**

579) The difference between TT and TdTd is 3° C. The atmosphere is

- a) Very Dry
- b) **Moist**
- c) Saturated

560) Just before the METAR was issued the weather was

- a) Rain
- b) Mist
- c) **Fog**

561) The range of pressure reported as Q1003 is

- a) 1002.5 to 1003.5 hPa
- b) **1003.0 to 1003.9 hPa**
- c) 1003.1 to 1003.5 hPa

562) The range of temperature reported as 32 is

- a) 31.5 to 32.4 ° C
- b) 32.1 to 32.4 ° C
- c) **31.5 to 32.4 ° C**

General Question on METAR and SPECI

563) Visibility is reported in steps of 50 m when visibility is

- a) 800 m to 5000 m
- b) 0 to 800 m**
- c) 5000 m to 10 km

564) Visibility is reported in steps of 100 m when visibility is

- a) 800 m to 5000 m**
- b) 0 to 800 m
- c) 5000 m to 10 km

565) Visibility is reported in steps of 1000 m when visibility is

- a) 800 m to 5000 m
- b) 0 to 800 m
- c) 5000 m to 9999**

566) Visibility is reported 9999 when visibility is

- a) 800 m to 5000 m
- b) 9000 to 9999 m
- c) 10 km or more**

567) Temperature + 2.5°C reported as

- a) 2°C
- b) 3°C**
- c) 2.5°C

568) Temperature - 12.5°C reported as

- a) - 12
- b) - 13
- c) M 12**

569) Pressure is rounded down to the nearest whole number in hectopascal

- a) exact value
- b) upper value
- c) lower value**

570) QNH 1002.6 hPa is reported as

- a) Q1002**
- b) Q1003
- c) Q1002.6

571) QNH 29.92 inches is reported as

- a) Q2900
- b) A2992**
- c) A 3000

572) Fog is reported when visibility is

- a) < 1000 m**
- b) 1000 m
- c) > 1000 m

573) Mist is reported when visibility is

- a) < 1000 m to 2000m
- b) 1000 to 5000m**
- c) >1000 m

574) CAVOK signifies Visibility, Cloud and present weather better than the prescribed values or conditions

- a) Ceiling**
- b) base
- c) amount

575) CAVOK is issued when visibility is

- a) 9-10 km
- b) 10 km or more**
- c) 10 km only

576) CAVOK is issued when (i) Visibility 10 km or more (ii) No weather of significance (iii) No clouds below 1500 m or below the highest minimum sector altitude, whichever is greater and no cumulonimbus.

- a) Any one condition
- b) Any two conditions
- c) All the three conditions**

577) SPECI is issued when

- (i) Change in wind direction by $\geq 60^\circ$ and speed before and/ after change is ≥ 10 kt.
- (ii) Mean speed has changed by ≥ 10 kt
- (iii) Variation from mean speed by ≥ 10 kt and speed before and/ after change ≥ 15 kt

- a) Any two conditions
- b) Any one condition**
- c) Any one condition

578) SPECI is issued when Clouds are

- (i) BKN or OVC base at 30, 60, 150, 300, 450 m
- (ii) Cloud amount below 450 m changes:

From SKC/ FEW/ SCT to BKN / OVC

From BKN/ OVC to SKC/ FEW/ SCT.

- a) Any one condition**
- b) Any two conditions
- c) All the three conditions

579) SPECI is issued for vertical visibility, by stations having Ceilograph, when sky is obscured

- a) True**
- b) False

580) SPECI is issued when surface temperature has increased by or more from the last observation.

- a) 3°C
- b) 4°C
- c) 2°C**

581) WSRWY28 indicates that the wind shear has been reported for RW 28 in

- a) Take off path
- b) approach path
- c) both take off and approach paths**

582) R26/M0150 in a METAR indicates that

- a) maximum RVR is 150 m
- b) minimum RVR is 150 m
- c) ***150 m is the lowest RVR instrument can measure and RVR is < 150m***

26. AVIATION WEATHER FORECASTS

(CODES OF AERODROME FORECAST, AREA FORECAST AND ROUTE)

1. AERODROME FORECAST (TAF)

TAF VILK 241800Z 2500/2509 09008KT 0800 FG BECMG 2504/2505 09015KT 6000 SCT008 BKN120 TEMPO 2506/2508 12015G30KT 3000 TSRA FEW012 FEW025CB BKN100 BECMG AT 25/0800 09010KT 7000 FEW030 SCT120 BKN280 =

583) The TAF has been issued on day

- a) 23rd
- b) 18lh
- c) ***24th***

584) The TAF has been issued at

- a) ***2330 IST***
- b) 1830 UTC
- c) 24 UTC

585) The expected surface wind speed is

- a) 09 kt
- b) 06 kt
- c) ***08 kt***

586) Initially expected wind direction is

- a) 120°
- b) ***090°***
- c) 050°

587) TAF is Valid for Date

- a) 23rd
- b) 24th
- c) ***25th***

588) Lowest forecast visibility in TAF is

- a) 1500m
- b) ***0800 m***
- c) 0200 m

589) Height of base of lowest clouds in TAP is

- a) 1000 m
- b) 1000 ft
- c) ***0800 ft***

590) Weather TSRA is expected after

- a) **0600 UTC**
- b) 0600 1ST
- c) 0800 UTC

591) Direction of gusty wind is

- a) 090°
- b) 100°
- c) **120°**

592) Expected weather up to 0400UTC is

- a) **Fog**
- b) TSRA
- c) Mist

593) Amount of lowest cloud is

- a) 1-2/8
- b) **3-4/8**
- c) 5-7/8

594) Amount of CB cloud is

- a) **1-2/8**
- b) 2-4/8
- c) 5-7/8

595) Height of base of CB is

- a) **2500 ft**
- b) 3000 m
- c) 2500 m

596) The height of topmost layer of cloud is

- a) 2800 m
- b) 28000 m
- c) **28000 ft**

597) Period of validity of TAF is

- a) 23 to 00 1ST
- b) **00 to 09 UTC**
- c) 20 to 12 UTC

Other Question

598) Group 1500/1509 in a TAF indicates

- a) TAF is issued on 15th at 0000 UTC
- b) **TAF is valid from 15th 0000 to 0900 UTC**
- c) TAF is valid from 1500 to 0900 UTC

599) What is true of a TAF

- a) 9 hr TAF is for international dissemination
- b) TAF valid for 9 hr is issued every 6 hr
- c) TAF valid for 12 - 30 hr is issued every 3 hr
- d) **TAF for national use are valid for 9 hr and issued every 3 hr**

2. ROUTE FORECAST FOR AVIATION (ROFOR)

**ROFOR 010000Z 010610 KT VECC VILK 2SC030 2CB030 3AC100 2CI300 7///170 621800 541501 405022
28015 407010 28020 410005 29030 420M05 27045 440M41 27105 11111 12870 380120 22222 36140 2825=**

600) Time of issue of ROFOR is